FYTA — Data Engineer assessment

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Three weeks, 8 soil sensors, ~16,000 readings, 49 user notes, 28 plant photos. `uv run fyta_pipeline.py` reproduces everything below.

Five things that matter

1. A sensor stopped measuring, and nothing says so. Sensor 6's moisture reads exactly 25.01 for fourteen days — 1,344 readings — and looks normal. What gives it away is that its *own* four other channels kept moving.

2. A two-day gap that leaves no trace. One sensor is missing 193 readings. No blank cells, no errors — the rows are simply absent. Only the expected-slot count reveals it.

3. A sensor in the wrong units, on the wrong clock. Sensor 7 read Fahrenheit and ran two hours out. Neither is visible in the values — Fahrenheit only from jittering 1.8× faster than its peers, the clock only from its daylight pattern.

4. One join is safe, the next is not. Plant IDs repeat, so joining notes or photos multiplies rows — 49 become 67. The trap is that the *first* join works.

5. Two pots have two sensors each, and the pairs disagree. Their moisture correlates at −0.08 and +0.14 — flat to negative. What that rules out is *averaging them*: the average describes neither pot and discards the disagreement. What it does **not** establish is why — depth, placement, a failed probe and a wrong mapping all produce this. **I could not settle it from the data alone.**

Three decisions I would defend

Not everything out of range is wrong. Impossible readings are removed and logged; merely unusual ones are *kept* and marked. Collapsing the two is how real events get deleted — one sensor has 431 unusual readings in long stretches.

Finding and fixing are separate jobs, checked both ways. A detector with no fix is a warning nobody acts on; a fix with no detector rots. The gate proves every removed value was removed, *and* every flag matches a detector.

Never average the two sensors on one pot — the disagreement is the only reason to fit two probes.

What I would never ship

- **Averaged sensors on one pot** — the disagreement *is* the signal.
- **A filled-in gap** — a user note sits inside those missing two days. An *expected-but-absent* row is honest; an interpolated one is not.
- **Trimmed readings** — 107 % moisture is a finding, not a typo to clip.
- **Deleted odd readings** — eight 1am light readings are out of pattern but inside range. Someone may have switched a lamp on. Kept, marked.
- **A safety check that cannot fail.**

Where I was wrong

Two external reviews found ten real defects *after* I had reported the work sound. All are fixed, each with a test that fails if the fix is undone. The four worst:

- **The publishing gate could not fail.** It asked "is any value missing?", which any missing value satisfies. I restored 1,344 known-bad readings; it passed.
- **Two rows on one key were resolved by coin-flip.** Disagreeing duplicates were dropped to one, silently. Both are now quarantined.
- **A falling reading counted as agreement** — a large conductivity *drop* scored as support for "fertiliser raised conductivity".
- **A frozen-channel rule deleted real data.** It nulled any long unchanged run, including legitimate darkness — 52 good light readings in a fixture.

One review also caught a claim that was simply wrong: I had written that a drying *rate* escapes the substrate calibration. It does not — differentiating removes the offset and keeps the gain, and the gain is what differs between soils.

The pattern: I checked by re-reading my own writing, not by re-running.

Two results that came back negative

The photo model cannot be validated here. Its water-deficiency score tracks sensor dryness at $r = 0.75$ — the one axis the sensor already measures — and the dataset is synthetic, so the agreement may be a shared generator rather than two instruments. The heads that *could* disagree (fungus, pests) never exceed 0.06. A head that never fires can neither confirm nor contradict.

Triangulation does not work, and not because the effect is weak. Across five fertilising events no probe's conductivity moves outside its own step noise — and the light logs *contradict*: "moved closer to window" moves the daily peak **down** by 4.4.

What I would do next

1. **Put the checks in the build** — every new check gets a test that fails without it.
2. **Move device settings into an effective-dated registry.** Unit fixes, clock offsets and specs are code today; plant, soil and species are already data.
3. **Settle the two-sensor question** — it decides how the modelling table is built.
4. **Version each derived value.** Moisture comes from a curve with no version.